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Introduction

Dr. Kailun Yang received his dual B.S. degrees in Measurement Technology and Instruments from the Beijing Institute of Technology (BIT) and Economics from Peking University (PKU) in June 2024. He received his Ph.D. degree in Information Sensing and Instrumentation with distinction from the State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University (ZJU) in June 2019. He performed a Ph.D. internship at the Robotics and eSafety (RobeSafe) research group at the University of Alcalá (UAH) from September 2017 to September 2018. He was a postdoctoral researcher with the Computer Vision for Human-Computer Interaction (CV:HCI) Lab at Karlsruhe Institute of Technology (KIT) from November 2019 to January 2023. He is currently a full professor at the School of Artificial Intelligence and Robotics and the National Engineering Research Center of Robot Visual Perception and Control Technology, Hunan University (HNU), where he directs Computer Vision for Panoramic Understanding (CV:PU) lab.

Dr. Kailun Yang has been researching computational imaging and computer vision for scene understanding in autonomous vehicles, human-computer interaction systems, and robotics. He has published more than 200 referred papers in journals including IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), International Journal of Computer Vision (IJCV), IEEE Transactions on Image Processing (TIP), IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Transactions on Intelligent Transportation Systems (T-ITS), IEEE Transactions on Multimedia (TMM), IEEE Transactions on Automation Science and Engineering (T-ASE), IEEE Transactions on Intelligent Vehicles (T-IV), IEEE Transactions on Instrumentation and Measurement (TIM), IEEE Transactions on Computational Imaging (TCI), IEEE Transactions on Artificial Intelligence (TAI), and top international conferences including CVPR, NeurIPS, ICCV ECCV, AAAI, IJCAI, CoRL ICRA, IROS, MM, IV, ITSC. He has been cited more than 9400 times with an H-Index of 48.

His work CMX was selected as a Top-10 Popular Article in IEEE Transactions on Intelligent Transportation Systems (T-ITS). His work Panoramic Panoptic Segmentation received a Best Paper Award at the 2021 IEEE Intelligent Vehicles Symposium (IV). His work MateRobot was selected as a Finalist for the Best Paper Award on Human-Robot Interaction at ICRA 2024. His work ACNet is listed as one of the Most Cited Papers in ICIP 2019. He received the Best Presentation Award at ICFIP 2018, Accessibility Challenge Judges' Award at W4A 2023, and Editors' Picks at Applied Optics [2018] [2019].

Dr. Kailun Yang serves as an associate editor for IEEE Transactions on Intelligent Transportation Systems, IEEE Robotics and Automation Letters, Robot Learning, IEEE ICRA, IEEE/RSJ IROS, and IEEE IV. He serves as an area chair for ICLR. He serves as an organizer for the 2nd Workshop on Multispectral Imaging for Robotics and Automation (MIRA) at the 2025 IEEE/CVF International Conference on Computer Vision (ICCV). He serves as an organizer for the 1st Workshop on Intelligent Cobodied Assistance and Robotic Empowerment (iCARE) at the Conference on Robot Learning (CoRL), 2025. He serves as an organizer for the 2nd SCSU-ITS: Synthetic-Data-Aided Safety-Critical Scenario Understanding in ITS at the 2025 IEEE International Conference on Intelligent Transportation Systems (ITSC). He serves as an organizer for the 2025 Chinese Congress on Image and Graphics (CCIG). He serves as an organizer and co-chair for the 2022 IEEE Vehicles Symposium (IV) Workshop on Beyond Supervised Learning: Addressing Data Scarcity in Intelligent Transportation Systems. He serves as the local chair for the 2024 International Symposium on Intelligent Robotics and Systems (ISoIRS). He was on the scientific committee for the 2020 International

Workshop of Physical Agents (WAF). He serves as a reviewer for more than 100 journals, including TPAMI, IJCV, TIP, T-ITS, and international conferences, including ICML, NeurIPS, ICLR, CVPR, ICCV, ECCV, ICRA, and IROS. He was selected as an outstanding reviewer for IEEE Transactions on Instrumentation and Measurement 2024, ECCV 2022, and ACCV 2022.

Research Interests

Computer Vision: Deep Learning, Semantic Segmentation, Panoptic Segmentation, Depth Estimation, Optical Flow Estimation, Dense Prediction, Knowledge Distillation, Attention Mechanisms, Vision Transformers, Unsupervised Learning, Omnidirectional Vision, Multi-modal Vision.
Intelligent Transportation Systems: Intelligent Vehicles, Autonomous Driving, Scene Understanding, Domain Adaptation, LiDAR Point Cloud Segmentation, Road Detection, BEV HD Mapping, V2X Collaborative Perception, Semantic Scene Completion, Semantic Occupancy Estimation, World Models.
Robotics: 3D Vision, Embodied Perception, Obstacle Avoidance, Sensor Fusion, Visual Localization, Visual Odometry, SLAM, Action Recognition, Affordance Reasoning, Human-Computer Interaction.
Optical Sensing: RGB-D and RGB-X Vision, Stereo Vision, Panoramic Imaging, Polarization Imaging, Computational Imaging, Event Cameras, Minimalist Optical Systems, Computational Aberration Correction.
Assistive Technologies: Advanced Driver-Assistance Systems (ADAS), Wearable Assistance Systems for People with Visual Impairments, Accessibility.

Experience & Education

2023.11 – Present **Full Professor – School of Artificial Intelligence and Robotics, Hunan University (HNU), China.** *Doctoral Advisor and Director of the Computer Vision for Panoramic Understanding (CV:PU) Lab.*

2023.2 – 2023. 10 **Associate Professor – School of Artificial Intelligence and Robotics, Hunan University (HNU), China.** *Doctoral Advisor.*

2019.11 – 2023.1 **PostDoctoral Researcher – Computer Vision for Human-Computer Interaction (CV:HCI) Lab, Karlsruhe Institute of Technology (KIT), Germany.** *Leading Computer Vision Research for Accessible Maps and Visually Impaired People.*

2019.07– 2019.10 **Research Assistant – National Engineering Research Center of Optical Instrumentation, Zhejiang University (ZJU), China.** *Collaborating in Vision and Robotics Research.*

2014.09– 2019.06 **PhD Candidate – State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University (ZJU), China.** *Majoring in Information Sensing and Instrumentation. PhD Project: 3D Terrain Perception for Navigation Assistance Systems.*

2017.09 – 2018.06 **Joint Training PhD Student – Robotics and eSafety (RobeSafe) Research Group, University of Alcalá (UAH), Spain.** *Project: Unified Semantic Perception.*

2010.09 – 2014.06 **Beijing Institute of Technology (BIT).** *Bachelor Degree in Measurement Technology and Instrumentation.*

2012.09 – 2014.06 **Peking University (PKU).** *Dual Degree in Economics.*

Main Awards

2025.01 **Outstanding Reviewer of 2024 for IEEE Transactions on Instrumentation and Measurement.**

2024.04 **Finalist for Best Paper Award on Human-Robot Interaction at IEEE ICRA 2024.**

2023.04 **Accessibility Challenge Judges' Award at W4A 2023.**

2022.12 **Outstanding Reviewer at ACCV 2022.**

2022.10 **Outstanding Reviewer at ECCV 2022.**

2021.07 **Best Paper Award at 2021 IEEE Intelligent Vehicles Symposium (IV).**

2019.06 **PhD Graduated with Distinction & Provincial Outstanding Graduate Student.**

2018.11 **National Scholarship for PhD Students.**

2018.10 **Excellent Paper Award at IET IRC 2018.**

2018.08 **Best Student Paper at 2018 SPIE Security+Defence Conference.**

2017.08 **"Youth Creation" China Youth Internet Entrepreneurship Competition, Gold Award & National Champion.**

2017.07 **Zhejiang Province "Internet +" Student Innovation and Entrepreneurship Competition, Golden Prize.**

2018.03 **Best Paper Award in ICFIP2018, Barcelona, Spain.**

2017.08 **China College Students Entrepreneurship Competition, Champion.**

2017.07 **"Internet+" Innovation and Entrepreneurship Competition, Golden Prize.**

2017.05 **"Challenge Cup" College Students Academic Competition, First Prize.**

Publications as 1st Author (22 Papers)

1. H. Liu*, J. Zhang*, **K. Yang***, X. Hu, R. Liu, R. Stiefelhagen. CMX: Cross-Modal Fusion for RGB-X Semantic Segmentation. **IEEE Transactions on Intelligent Transportation Systems**, 2023. **Top-10 Popular Articles in T-ITS.**
2. Z. Wang*, **K. Yang***, H Shi, P Li, F Gao, J Bai, K Wang. LF-VISLAM: A SLAM Framework for Large Field-of-View Cameras with Negative Imaging Plane on Mobile Agents. **IEEE Transactions on Automation Science and Engineering**, 2023.
3. **K. Yang**, X. Hu, R. Stiefelhagen. Is Context-Aware CNN Ready for the Surroundings? Panoramic Semantic Segmentation in the Wild. **IEEE Transactions on Image Processing**, 2021.
4. **K. Yang**, J. Zhang, S. Reiß, X. Hu, R. Stiefelhagen. Capturing Omni-Range Context for Omnidirectional Segmentation. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2021.
5. **K. Yang**, X. Hu, Y. Fang, K. Wang, R. Stiefelhagen. Omnisupervised Omnidirectional Semantic Segmentation. **IEEE Transactions on Intelligent Transportation Systems**, 2020.
6. **K. Yang**, X. Hu, K. Wang, R. Stiefelhagen. In Defense of Multi-Source Omni-Supervised Efficient ConvNet for Robust Semantic Segmentation in Heterogeneous Unseen Domains. In IEEE Intelligent Vehicles Symposium (**IV**), 2020.
7. **K. Yang**, X. Hu, H. Chen, K. Xiang, K. Wang, R. Stiefelhagen. DS-PASS: Detail-Sensitive Panoramic Annular Semantic Segmentation through SwaftNet for Surrounding Sensing. In IEEE Intelligent Vehicles Symposium (**IV**), 2020.
8. **K. Yang**, X. Hu, L.M. Bergasa, E. Romera, K. Wang. PASS: Panoramic Annular Semantic Segmentation. **IEEE Transactions on Intelligent Transportation Systems**, 2019.
9. **K. Yang**, L.M. Bergasa, E. Romera, K. Wang. Robustifying Semantic Cognition of Traversability across Wearable RGB-Depth Cameras. **Applied Optics**, 2019. **Highlighted as an Editors' Pick.**
10. **K. Yang**, X. Hu, L.M. Bergasa, E. Romera, X. Huang, D. Sun, K. Wang. Can we PASS beyond the Field of View? Panoramic Annular Semantic Segmentation for Real-World Surrounding Perception. In IEEE Intelligent Vehicles Symposium (**IV**), 2019.
11. **K. Yang**, K. Wang, L.M. Beragsa, E. Romera, W. Hu, D. Sun, J. Sun, R. Cheng, T. Chen, E. López. Unifying Terrain Awareness for the Visually Impaired through Real-Time Semantic Segmentation. **Sensors**, 2018.
12. **K. Yang**, K. Wang, H. Chen, J. Bai. Reducing the minimum range of a RGB-depth sensor to aid navigation in visually impaired individuals. **Applied Optics**, 2018. **Highlighted as an Editors' Pick.**
13. **K. Yang**, R. Cheng, L.M. Bergasa, E. Romera, K. Wang, N. Long. Intersection perception through real-time semantic segmentation to assist navigation of visually impaired pedestrians. In IEEE International Conference on Robotics and Biomimetics (**ROBIO**), 2018.
14. **K. Yang**, L.M. Beragsa, E. Romera, J. Wang, K. Wang, E. López. Perception framework of water hazards beyond traversability for real-world navigation assistance systems. In IEEE International Conference on Robotics and Biomimetics (**ROBIO**), 2018.

15. **K. Yang**, L.M. Beragsa, E. Romera, X. Huang, K. Wang. Predicting Polarization Beyond Semantics for Wearable Robotics. In IEEE-RAS International Conference on Humanoid Robots (**Humanoids**), 2018.
16. **K. Yang**, L.M. Bergasa, E. Romera, D. Sun, K. Wang, R. Barea. Semantic perception of curbs beyond traversability for real-world navigation assistance systems. In IEEE International Conference on Vehicular Electronics and Safety (**ICVES**), 2018.
17. **K. Yang**, L.M. Bergasa, E. Romera, R. Cheng, T. Chen, K. Wang. Unifying terrain awareness through real-time semantic segmentation. In IEEE Intelligent Vehicles Symposium (**IV**), 2018.
18. **K. Yang**, K. Wang, S. Lin, J. Bai, L.M. Bergasa, R. Arroyo. Long-Range Traversability Awareness and Low-Lying Obstacle Negotiation with RealSense for the Visually Impaired. In International Conference on Frontiers of Image Processing (**ICFIP**), 2018. **Best Presentation Award.**
19. **K. Yang**, K. Wang, R. Cheng, W. Hu, X. Huang, J. Bai. Detecting Traversable Area and Water Hazards for the Visually Impaired with a pRGB-D Sensor. **Sensors**, 2017.
20. **K. Yang**, K. Wang, X. Zhao, R. Cheng, J. Bai, Y. Yang, D. Liu. IR Stereo RealSense: Decreasing minimum range of navigational assistance for visually impaired individuals. **Journal of Ambient Intelligence and Smart Environments**, 2017.
21. **K. Yang**, K. Wang, W. Hu, J. Bai. Expanding the Detection of Traversable Area with RealSense for the Visually Impaired. **Sensors**, 2016.
22. **K. Yang**, K. Wang, R. Cheng, X. Zhu. A new approach of point cloud processing and scene segmentation for guiding the visually impaired. In IET International Conference on Biomedical Image and Signal Processing (**ICBISP**), 2015.

Publications as Corresponding Author (88 Papers)

1. J. Wei, Z. Teng, F. Teng, J. Zheng, R. Liu, Y. Chen, J. Hu, **K. Yang**†, J. Zhang†, R. Stiefelhagen. OneBEV++: Towards Unifying Bird's-Eye-View Semantic Mapping with Panoramas. **IEEE Transactions on Pattern Analysis and Machine Intelligence**, 2026.
2. K. Peng†, D. Wen, M.S. Sarfraz, Y. Chen, J. Zheng, D. Schneider, **K. Yang**†, J. Wu, A. Roitberg, R. Stiefelhagen. Mitigating Label Noise using Prompt-Based Hyperbolic Meta-Learning in Open-Set Domain Generalization. **International Journal of Computer Vision**, 2026.
3. S. Li, F. Teng, Y. Cao, **K. Yang**†, Z. Li†, Y. Wang. NRSeg: Noise-Resilient Learning for BEV Semantic Segmentation via Driving World Models. **IEEE Transactions on Image Processing**, 2026.
4. S. Li, Y. Cao, H. Shi, Y. Zang, X. He, **K. Yang**†, Z. Li†. HierDAMap: Toward Universal Domain Adaptive BEV Mapping via Hierarchical Perspective Priors. **IEEE Transactions on Intelligent Transportation Systems**, 2026.
5. M. Duan, Y. Zhang, Y. Cao, F. Teng, K. Luo, J. Zhang, **K. Yang**†, Z. Li†. Panoramic Out-of-Distribution Segmentation. **IEEE Transactions on Circuits and Systems for Video Technology**, 2026.
6. Z. Yi, H. Shi, Q. Jiang, Y. Gao, Z. Wang, Y. Zhang, **K. Yang**†, K. Wang†. Towards Dependable Optical Flow: Benchmarks for Real-World Spatiotemporal Corruptions. **IEEE Transactions on Circuits and Systems for Video Technology**, 2026.
7. S. Li, **K. Yang**†, H. Shi, S. Wang, Y. Yao, Z. Li†. GenMapping: Unveiling the Potential of Inverse Perspective Mapping for Robust Online HD Map Construction. **IEEE Open Journal of Intelligent Transportation Systems**, 2026.
8. Y. Qin, S. Fan, F. Yang, Z. Xue, Z. Mai, W. Chen, **K. Yang**†, Z. Li†. Learning Fine-Grained Correspondence with Cross-Perspective Perception for Open-Vocabulary 6D Object Pose Estimation. **IEEE Robotics and Automation Letters**, 2026.
9. B. Deng, L. Huang, K. Luo, F. Teng, **K. Yang**†. DepTR-MOT: Unveiling the Potential of Depth-Informed Trajectory Refinement for Multi-Object Tracking. In IEEE International Conference on Robotics and Biomimetics (**ROBIO**), 2026.
10. K. Peng†, Z. Zhou, **K. Yang**†, D. Wen, R. Liu, Y. Chen, J. Zheng, H. Shi, Y. Zhou, M.S. Sarfraz, D.P. Paudel, L. Van Gool. Seeing Together: Multi-Robot Cooperative Egocentric Spatial Reasoning with Multimodal Large Language Models. In Conference on Neural Information Processing Systems (NeurIPS), 2026.
11. Y. Lin, K. Luo, Y. Zheng, H. Shi, M. Duan, Y. Liu, **K. Yang**†. OccTrack360: 4D Panoptic Occupancy

Tracking from Surround-View Fisheye Cameras. In IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2026.

12. G. Zhu, W. Jia, C. Shao, Y. Zhang, Z. Li, **K. Yang**†. PanoAffordanceNet: Towards Holistic Affordance Grounding in 360° Indoor Environments. In IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2026.
13. Z. Yang, G. Zhao, S. Wu, K. Luo, **K. Yang**†. Spherical-GOF: Geometry-Aware Panoramic Gaussian Opacity Fields for 3D Scene Reconstruction. In IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2026.
14. K. Luo, X. Wang, R. Fan, **K. Yang**†. NOVA: Next-step Open-Vocabulary Autoregression for 3D Multi-Object Tracking in Autonomous Driving. In IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2026.
15. W. Li, K. Peng†, D. Wen, J. Zheng, J. Wei, M. Duan, Y. Zhang, R. Fan, **K. Yang**†. Can we Trust Unreliable Voxels? Exploring 3D Semantic Occupancy Prediction under Label Noise. In IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2026.
16. K. Luo, F. Teng, M. Duan, W. Jia, X. Wang, H. Shi, K. Peng, Z. Li, **K. Yang**†. PS-MOT: Cultivating Instance Awareness from Point Seeds for Multi-object Tracking. In European Conference on Computer Vision (**ECCV**), 2026.
17. R. Liu, Y. Chen, Y. Zhang, J. Zheng, K. Peng, C. Wu, C. Huang, D. Wen, J. Zhang, **K. Yang**†, R. Stiefelhagen. What If? Emulative Simulation with World Models for Situated Reasoning. In European Conference on Computer Vision (**ECCV**), 2026.
18. Y. Zheng, K. Peng, X. Zheng, **K. Yang**†. Seeing Beyond: Extrapolative Domain Adaptive Panoramic Segmentation. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
19. Y. Zhang, M. Duan, K. Peng†, Y. Wang, D. Wen, D.P. Paudel, L. Van Gool, **K. Yang**†. ProOOD: Prototype-Guided Out-of-Distribution 3D Occupancy Prediction. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
20. H. Shi, Z. Wang, S. Guo, M. Duan, S. Wang, T. Chen, **K. Yang**†, L. Wang†, K. Wang†. OneOcc: Semantic Occupancy Prediction for Legged Robots with a Single Panoramic Camera. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
21. F. Teng, K. Luo, S. Wu, S. Li, P. Guo, J. Wei, J. Zhang, K. Peng, **K. Yang**†. Hallucinating 360°: Panoramic Street-View Generation via Local Scenes Diffusion and Probabilistic Prompting. In IEEE International Conference on Robotics and Automation (**ICRA**), 2026.
22. W. Li, K. Peng†, D. Wen, R. Liu, M. Duan, K. Luo, **K. Yang**†. Segment-to-Act: Label-Noise-Robust Action-Prompted Video Segmentation Towards Embodied Intelligence. In IEEE International Conference on Robotics and Automation (**ICRA**), 2026.
23. L. Kong, J. Lin, S. Li, K. Luo, Z. Li†, **K. Yang**. CoBEVMoE: Heterogeneity-aware Feature Fusion with Dynamic Mixture-of-Experts for Collaborative Perception. In IEEE International Conference on Robotics and Automation (**ICRA**), 2026.
24. S. Guo, H. Shi, S. Wang, X. Yin, **K. Yang**†, K. Wang†. Event-Aided Semantic Scene Completion. In IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), 2026.
25. J. Sun, W. Sun†, G. Zhang†, **K. Yang**†, S. Li, X. Meng, N. Deng, C. Tan. CT-UIO: Continuous-Time UWB-Inertial-Odometer Localization Using Non-Uniform B-spline with Fewer Anchors. **IEEE Transactions on Mobile Computing**, 2025.
26. H. Shi, S. Wang, J. Zhang, X. Yin, G. Wang, J. Zhu, **K. Yang**†, K. Wang†. Offboard Occupancy Refinement with Hybrid Propagation for Autonomous Driving. **IEEE Transactions on Intelligent Transportation Systems**, 2025.
27. H. Li, Q. Hu, B. Zhou, Y. Yao, J. Lin, **K. Yang**†, P. Chen†. CFMW: Cross-modality Fusion Mamba for Robust Object Detection under Adverse Weather. **IEEE Transactions on Circuits and Systems for Video Technology**, 2025.
28. X. Qian, Q. Jiang, Y. Gao, S. Gao, Z. Yi, L. Sun, K. Wei, H. Li, **K. Yang**†, K. Wang†, J. Bai. Towards Single-Lens Controllable Depth-of-Field Imaging via Depth-Aware Point Spread Functions. **IEEE Transactions on Computational Imaging**, 2025.
29. J. Chen, J. Lin, G. Zhong, H. Fu, K. Nai, **K. Yang**†, Z. Li†. Expression Prompt Collaboration Transformer for universal referring video object segmentation. **Knowledge-Based Systems**, 2025.

30. X. Yin, H. Shi, Y. Bao, Z. Bing, Y. Liao, **K. Yang**†, K. Wang†. E-3DGS: Gaussian Splatting with Exposure and Motion Events. **Applied Optics**, 2025.
31. Y. Ye, H. Shi, **K. Yang**†, Z. Wang, X. Yin, L. Sun, Y. Wang, K. Wang†. Towards Anytime Optical Flow Estimation with Event Cameras. **Sensors**, 2025.
32. Y. Cao, J. Zhang, X. Zheng, H. Shi, K. Peng, H. Liu, **K. Yang**†, H. Zhang†. Unlocking Constraints: Source-Free Occlusion-Aware Seamless Segmentation. In IEEE/CVF International Conference on Computer Vision (**ICCV**), 2025.
33. Y. Huang, F. Yang, G. Zhu, G. Li, H. Shi, Y. Zuo, W. Chen, Z. Li†, **K. Yang**†. Resource-Efficient Affordance Grounding with Complementary Depth and Semantic Prompts. In IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2025.
34. J. Zhao, F. Teng, K. Luo, G. Zhao, Z. Li, X. Zheng†, **K. Yang**†. Unveiling the Potential of Segment Anything Model 2 for RGB-Thermal Semantic Segmentation with Language Guidance. In IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2025.
35. Z. Wang, Y. Li, L. Xu, H. Shi, Z. Ma, Z. Zhu, C. Li, F. Gao, **K. Yang**†, K. Wang†. SF-TIM: A Simple Framework for Enhancing Quadrupedal Robot Jumping Agility by Combining Terrain Imagination and Measurement. In IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2025.
36. W. Jia, F. Yang, M. Duan, X. Chen, Y. Wang, Y. Jiang, W. Chen, **K. Yang**†, Z. Li†. One-Shot Affordance Grounding of Deformable Objects in Egocentric Organizing Scenes. In IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2025.
37. S. Wu, F. Teng, H. Shi, Q. Jiang, K. Luo, K. Wang, **K. Yang**†. QuaDreamer: Controllable Panoramic Video Generation for Quadruped Robots. In Conference on Robot Learning (**CoRL**), 2025.
38. K. Luo, H. Shi, S. Wu, F. Teng, M. Duan, C. Huang, Y. Wang, K. Wang, **K. Yang**†. Omnidirectional Multi-Object Tracking. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.
39. Z. Yi, H. Shi, Q. Jiang, **K Yang**†, Z Wang, D Gu, Y Zhang, K Wang†. EI-Nexus: Towards Unmediated and Flexible Inter-Modality Local Feature Extraction and Matching for Event-Image Data. In IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), 2025.
40. J. Zhang, **K. Yang**†, H. Shi, S. Reiß, K. Peng, C. Ma, H. Fu, F.H.S. Torr, K. Wang, R. Stiefelhagen. Behind Every Domain There is a Shift: Adapting Distortion-aware Vision Transformers for Panoramic Semantic Segmentation. **IEEE Transactions on Pattern Analysis and Machine Intelligence**, 2024.
41. Q. Jiang, S. Gao, Y. Gao, **K. Yang**†, Z. Yi, H. Shi, L. Sun, K. Wang†. Minimalist and High-Quality Panoramic Imaging With PSF-Aware Transformers. **IEEE Transactions on Image Processing**, 2024.
42. H. Shi, C. Pang, J. Zhang, **K. Yang**†, Y. Wu, H. Ni, Y. Lin, R. Stiefelhagen, K. Wang†. CoBEV: Elevating Roadside 3D Object Detection With Depth and Height Complementarity. **IEEE Transactions on Image Processing**, 2024.
43. J. Lin, J. Chen, **K. Yang**†, A. Roitberg, S. Li, Z. Li†, S. Li. AdaptiveClick: Click-Aware Transformer With Adaptive Focal Loss for Interactive Image Segmentation. **IEEE Transactions on Neural Networks and Learning Systems**, 2024.
44. R. Liu, **K. Yang**†, A. Roitberg, J. Zhang, K. Peng, H. Liu, Y. Wang, R. Stiefelhagen. TransKD: Transformer Knowledge Distillation for Efficient Semantic Segmentation. **IEEE Transactions on Intelligent Transportation Systems**, 2024.
45. S. Li, J. Lin, H. Shi, J. Zhang, S. Wang, Y. Yao, Z Li†, **K Yang**†. DTCLMapper: Dual Temporal Consistent Learning for Vectorized HD Map Construction. **IEEE Transactions on Intelligent Transportation Systems**, 2024.
46. J. Lin, J. Chen, K. Peng, X. He, Z. Li†, R. Stiefelhagen, **K. Yang**†. EchoTrack: Auditory Referring Multi-Object Tracking for Autonomous Driving. **IEEE Transactions on Intelligent Transportation Systems**, 2024.
47. Z. Wang, **K. Yang**†, H. Shi, Y. Zhang, Z. Xu, F. Gao, K. Wang†. LF-PGVIO: A Visual-Inertial-Odometry Framework for Large Field-of-View Cameras using Points and Geodesic Segments. **IEEE Transactions on Intelligent Vehicles**, 2024.
48. H. Shi, Q. Jiang, **K. Yang**†, X. Yin, Z. Wang, K. Wang†. Beyond the Field-of-View: Enhancing Scene Visibility and Perception with Clip-Recurrent Transformer. **IEEE Transactions on Intelligent Vehicles**, 2024.

49. Q. Jiang, H. Shi, S. Gao, J. Zhang, **K. Yang**†, L. Sun, H. Ni, K. Wang†. Computational Imaging for Machine Perception: Transferring Semantic Segmentation Beyond Aberrations. **IEEE Transactions on Computational Imaging**, 2024.
50. F. Teng, J. Zhang, K. Peng, Y. Wang, R. Stiefelhagen, **K. Yang**†. OAFuser: Towards Omni-Aperture Fusion for Light Field Semantic Segmentation. **IEEE Transactions on Artificial Intelligence**, 2024.
51. X. Yin, H. Shi, J. Chen, Z. Wang, Y. Ye, **K. Yang**†, K. Wang†. Exploring event-based human pose estimation with 3D event representations. **Computer Vision and Image Understanding**, 2024.
52. Q. Jiang, Z. Yi, S. Gao, Y. Gao, X. Qian, H. Shi, L. Sun, J. Niu, K. Wang†, **K. Yang**†, J. Bai. Representing domain-mixing optical degradation for real-world Computational Aberration Correction via vector quantization. **Optics & Laser Technology**, 2024.
53. K. Peng, D. Wen, **K. Yang**†, A. Luo, Y. Chen, J. Fu, M.S. Sarfraz, A. Roitberg, R. Stiefelhagen. Advancing Open-Set Domain Generalization Using Evidential Bi-Level Hardest Domain Scheduler. In Advances in Neural Information Processing Systems (**NeurIPS**), 2024.
54. F. Teng, J. Zhang, J. Liu, K. Peng, X. Cheng, Z. Li, **K. Yang**†. LF Tracy: A Unified Single-Pipeline Approach for Salient Object Detection in Light Field Cameras. In International Conference on Pattern Recognition (**ICPR**), 2024.
55. K. Zeng, H. Shi, J. Lin, S. Li, J. Cheng, K. Wang, Z. Li†, **K. Yang**†. MambaMOS: LiDAR-based 3D Moving Object Segmentation with Motion-aware State Space Model. In ACM International Conference on Multimedia (**MM**), 2024.
56. K. Peng, D. Schneider, A. Roitberg, **K. Yang**†, J. Zhang, C. Deng, K. Zhang, M.S. Sarfraz, R. Stiefelhagen. Towards Video-based Activated Muscle Group Estimation in the Wild. In ACM International Conference on Multimedia (**MM**), 2024.
57. K. Peng, J. Fu, **K. Yang**†, D. Wen, Y. Chen, R. Liu, J. Zheng, J. Zhang, M.S. Sarfraz, R. Stiefelhagen, A. Roitberg. Referring Atomic Video Action Recognition. In European Conference on Computer Vision (**ECCV**), 2024.
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